

CSCI 611 A4: Style Transfer Report

Repository Link: https://github.com/FlansaasSoft/CSCI611_Logan_Flansaas

Implementation

For this assignment I completed the image style transfer pipeline using the feature extractor portion of VGG19 given in the Jupyter notebook. Content loss was measured by comparing activations from a selected VGG layer between the target and content images, while style loss was measured by comparing Gram matrices between the target and style images across multiple convolutional layers. The target image was initialized from the content image and then optimized with Adam for 2000 steps while showing the progress of the image every 400 steps.

For my experiments I varied four main hyperparameters: the style-layer weighting profile, the global style weight, the learning rate, and the content layer. In total, 72 experiments were run by testing four layer-weighting profiles (early, middle, late, and equal), three style weights (10^3 , 10^6 , and 10^9), three learning rates (0.001, 0.003, and 0.01), and two content layers (conv2_2 and conv4_2). The final target images of each experiment were saved separately and screenshots were taken of the loss reports of each experiment. I tried to save all of the results in the Jupyter notebook, but the experiments made the notebook very large and attempts to save the notebook just hanged forever, so the intermediate results and side by side comparisons of each experiment were lost and the evidence in the notebook is limited.

Findings

Across all runs, the content image's main structure remained largely recognizable, but the wildfire style image strongly influenced the final color palette and texture, especially through bright orange tones and branch-like or ember-like surface patterns. Total loss also steadily decreased across all configurations, demonstrating stable convergence. The most noticeable differences came from the learning rate. Low learning rate runs preserved the original content the most. Facial features, edges, and the general structure of the subject remained clear, and the style appeared more like an overlay than a transformation. Medium learning rate runs created a stronger balance between recognizable content and visible stylization. High learning rate runs changed the target the most, pushing the results toward a more abstract and painted appearance. In these outputs, the texture from the style image often dominated the subject, and fine content details were more likely to be lost. Overall, learning rate had the strongest overall effect on how dramatic the style transfer looked.

The style-layer weighting profile also had a major impact on the final appearance. Runs that emphasized earlier VGG layers produced stronger local textures and more intense high frequency patterns, which made the outputs look more stylized but less faithful to the original content image. In contrast, runs that emphasized later layers tended to preserve the overall form of the content image much better. These images still reflected the color and texture of the style image, but the original content was more dominant overall. The middle and equal profiles usually fell somewhere between those two extremes. In general, earlier style-layer emphasis led to stronger texture transfer, while later style-layer emphasis led to better structure preservation.

The global style weight and the selected content layer had a much smaller effect than expected. Even though the style weight values ranged from 10^3 to 10^9 , many outputs within the same learning-rate and layer-profile group looked basically identical. This suggests that once the style term became large enough, other factors played a bigger role in shaping the final result. Changing the content layer from conv2_2 to conv4_2 also did not create significant visual differences in general, unlike in the VGG19 paper. There were some subtle shifts in how much local detail was preserved, but these changes were much less obvious than the differences caused by learning rate or layer-weighting strategy. A possible explanation for this is that all of the global style weights were sufficiently large to create strong stylization, and the number of steps was great enough that the system converged to very similar solutions in the end anyways.

Overall, the experiments showed that VGG-based image style transfer is highly sensitive to optimization choices, especially learning rate and the distribution of style loss across shallow versus deep layers. Higher learning rates and stronger emphasis on early style layers produced the most dramatic and abstract images, while lower learning rates and later style-layer emphasis produced more balanced results that better preserved the content image. Based on the results, the best balance between style and content came from configurations such as middle or late layer weighting combined with low or medium learning rates. The results demonstrate that relatively small changes in hyperparameters can significantly affect the visual quality and character of style transfer outputs.